

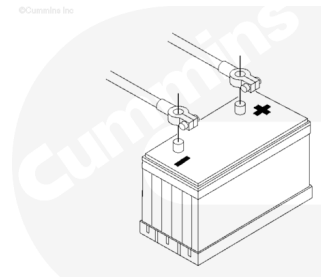
Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries.



- Remove the driven accessory. Refer to the OEM instructions

Note : Typically the driven accessory is a hydraulic pump.

- Refer to Procedure 009-016 (</qs3/pubsys2/xml/en/procedures/40/40-009-016.html>) for general removal instructions.

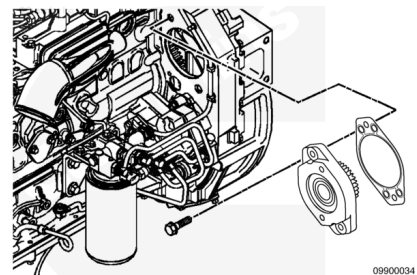


Remove

Remove the two capscrews securing the accessory drive to the rear gear housing.

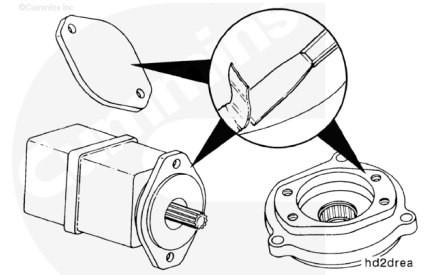
Remove the accessory drive and gasket.

Note : For front gear train engines it may be necessary to remove the external oil supply tube.



Clean and Inspect for Reuse

Remove any residual gasket material from the mounting surfaces of the accessory drive, driven accessory/hydraulic pump, and, if equipped, the cover plate surface.



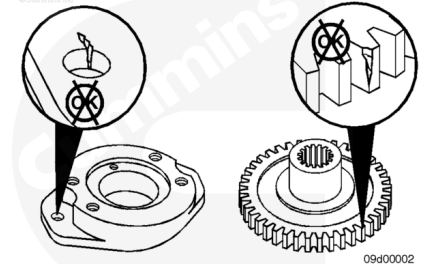
Note : If equipped, make sure the oil supply hole in the accessory drive housing is open and free of debris.

Inspect the accessory drive housing for cracks.

Inspect the accessory drive gear and shaft splines for cracks, broken teeth, and other damage.

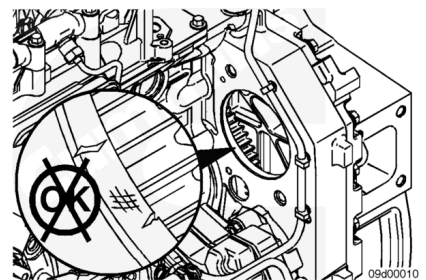
Note : If there is damage to the accessory drive gear teeth or there are signs of excessive heat, make sure to inspect the associated camshaft and fuel pump gears for damage. Measure camshaft gear backlash upon installation.

Replace any damaged components. See the Disassemble step of this procedure.



Note : For rear gear train engines, make sure the oil supply hole in the gear housing is open and free of debris.

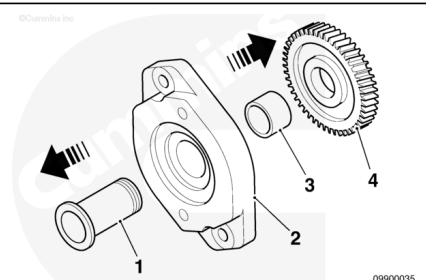
Inspect the inside diameter of the drive gear bore for excessive wear or damage; replace if necessary.



Disassemble

If damage was found to a component of the accessory drive during the Clean and Inspect for Reuse step, the following components of the accessory drive, if damaged, may be replaced by disassembling the accessory drive.

1. Accessory Drive Shaft
2. Accessory Drive Housing



3. Accessory Drive Bearing
4. Accessory Drive Gear.

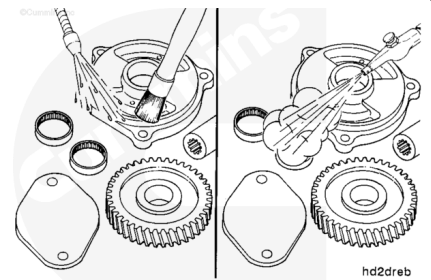
Note : Before disassembling the accessory drive, check to make sure the parts for the specific accessory drive being serviced are available. If not, replace the accessory drive as an assembly.

Use a hydraulic press and proper support to disassemble the accessory drive in the following order.

1. Press the drive shaft from the drive gear and housing
2. Press the bearing from the housing.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

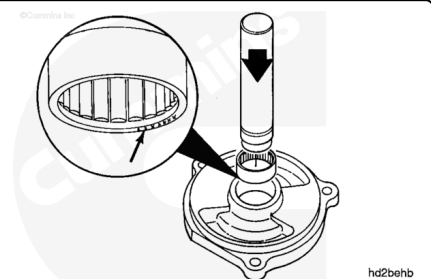
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the accessory drive components with solvent.

Dry with compressed air.

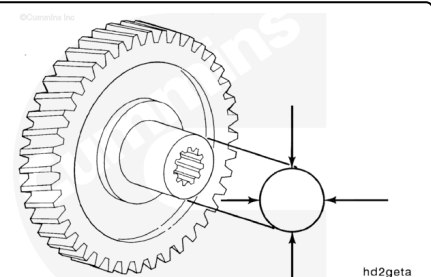
Inspect the bearing for binding, seizing, and excessive noise when spun.

Replace if any damage is found.



Measure the inside diameter of the accessory drive gear bore.

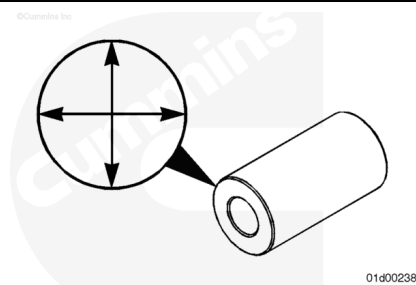
mm		in
38.920	MIN	1.5323
38.945	MAX	1.5333



If out of specification, replace the accessory drive gear.

Measure the outside diameter of the accessory drive shaft at multiple locations along the shaft.

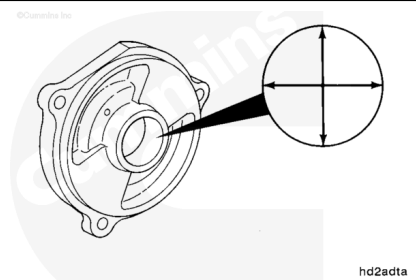
mm		in
39.008	MIN	1.5357
39.020	MAX	1.5362



If out of specification, replace the accessory drive shaft.

Measure the inside diameter of the bearing bore in the accessory drive adapter housing.

mm		in
67.759	MIN	2.6755
67.983	MAX	2.6765

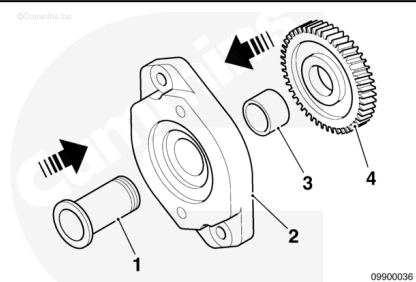


If out of specification, replace the housing.

Assemble

Use a hydraulic press and proper support to assemble the accessory drive in the following order:

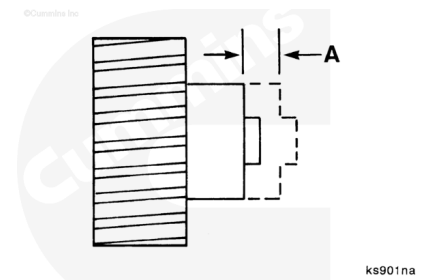
1. Press the accessory drive shaft into the bearing, while supporting the bearing inner race, until the drive shaft bottoms on the inner race
2. Press the outer race of the bearing, with drive shaft, into the accessory drive housing, until the outer race bottoms in the housing
3. Press the accessory drive gear onto the shaft while supporting the bottom of the drive shaft. Press until gear bottoms against inner bearing race.



Measure

Using gauge, Part Number 3824564, and magnetic base, Part Number 3377399, check the accessory drive gear end play.

mm		in
0.5	MIN	0.020



If the end plays is out of specification:

1. If just assembled, check if the drive gear and bearing were completely pressed onto the drive shaft
2. If **not** previously disassembled, disassemble the accessory drive and inspect for damage. Replace as necessary
3. Replace the accessory drive assembly.

Initial Check

Rear Gear Train

Note : If oil supply to the accessory drive is **not** required and the gasket does **not** have oil passages, this check is **not** required. If the accessory drive uses o-rings for seals, this check is **not** required.

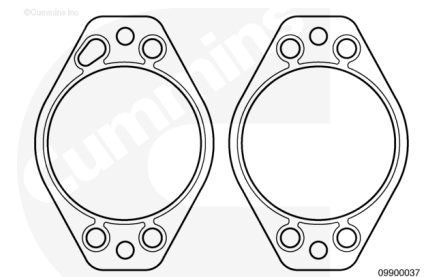
Prior to installing the accessory drive, identify which gasket is going to be installed so that, if necessary, the gasket can be properly oriented.

There are two types of accessory drive gasket:

1. Three round oil supply passages and one elongated oil supply passage
2. Four round oil supply passages.

It is preferred that, when installing the accessory drive, the gasket with the four round oil supply passages be used. The gasket can be installed in any orientation.

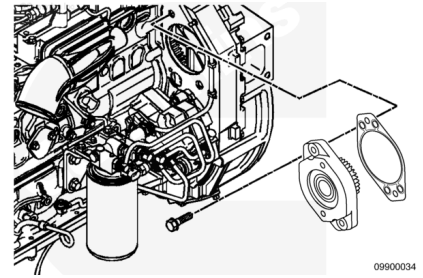
If only the gasket with the one elongated oil supply passage is available, install the gasket so that the elongated oil supply passage is **not** over the oil supply hole in the gear housing.



Install

⚠ CAUTION ⚠

Failure to line up the oil supply hole to the accessory drive properly will result in accessory drive damage.



Install the accessory drive and new gasket.

Note : If required, when installing the accessory drive and gasket, make sure the oil supply hole in the gear housing is lined up with the holes in the accessory drive and gasket. The accessory drive is marked for "Top" and "Bottom".

Install and tighten the two capscrews securing the accessory drive to the rear gear housing.

Torque Value: 62 n•m [46 ft-lb]

Finishing Steps

- Install the driven accessory. Refer to the OEM instructions.

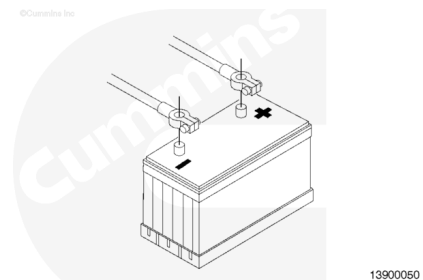
Note : Typically the driven accessory is a hydraulic pump.

- Refer to Procedure 009-016 (</qs3/pubsys2/xml/en/procedures/40/40-009-016.html>) for general installation instructions.



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- Connect the batteries
- Operate the engine and check for leaks.

